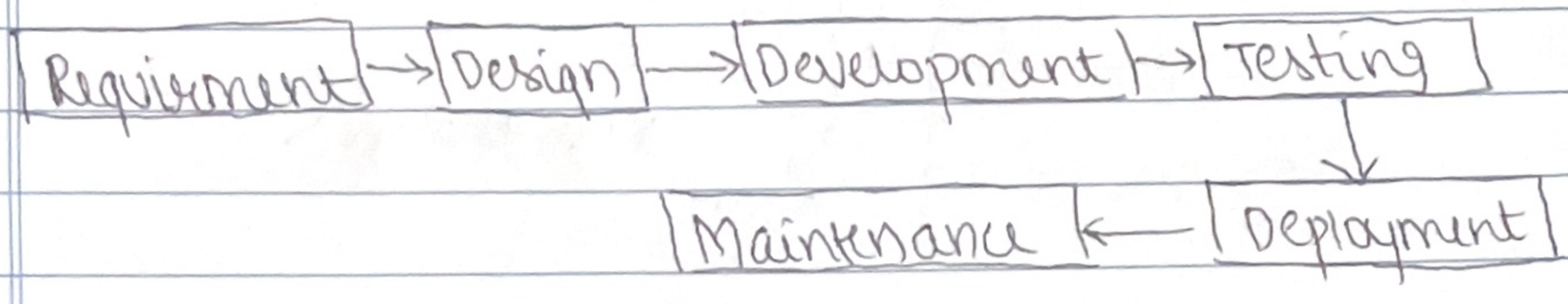
Garv Lalla
T1-3
59
SEPM

THADOMAL SHAHANI TSEC ENGINEERING COLLEGE

Software Process Models.

1] Waterfall Model



Waterfall model is one of the earliest and most traditional software process models. It follows linear & sequential approach.

Phases of waterfall model:

1] Requirment Analysis:
Gather & document all software requirments.
Output: software requirment specification.

2] System design:
Design system architecture, data flow & interfaces.
Output: Design documents (HLD & LLD)

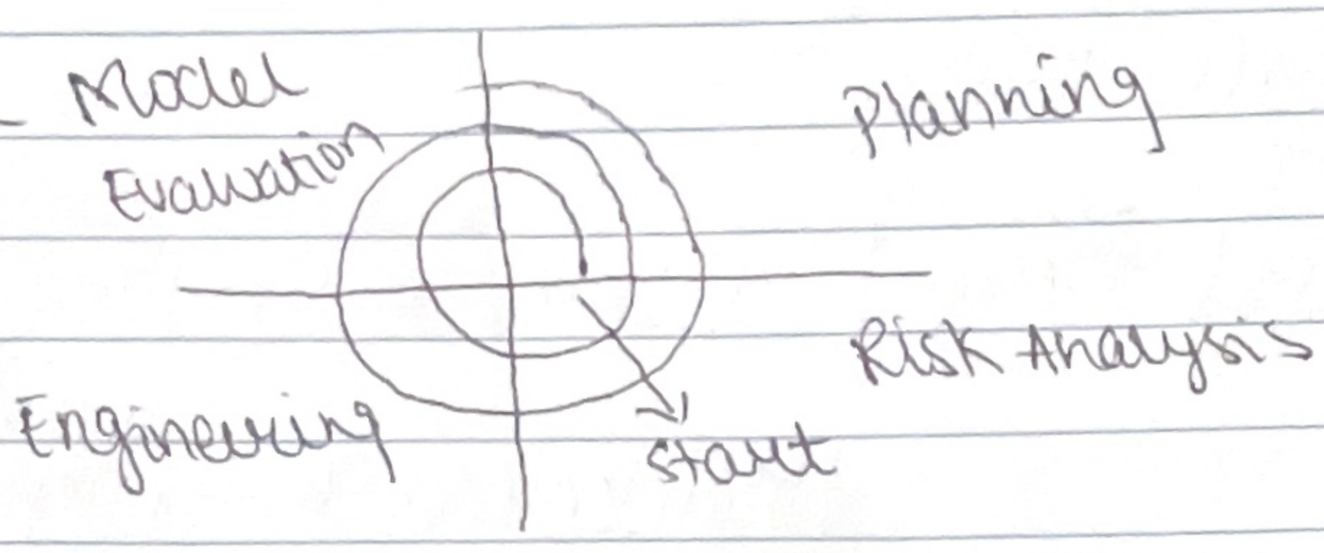
3] Development (coding):
convert design into source code.
Each module is developed and unit tested.

4] Testing:
Integrate modules and test the complete system.
Identify and fix defects.

5] Deployment:
Deliver any install the software for users.

6] Maintenance:
Fix bugs, improve performance, and update features.

2] Spiral Model



spiral model is a risk driven software process that combines iterative development with systematic risk analysis.
Development proceeds in a serial of loops.

Phases of the spiral model:
1] Planning:
Define objectives, alternatives and constraints.
Identify requirements for the iteration.
2] Risk Analysis:
Identify technical, schedule and cost risks.
Build prototypes to reduce risks.
3] Engineering:
Design, code and test the selected solution.
4] Evaluation:
Customer reviews the model.
Decide continue to the next spiral.

Each loop represents one phase of the software process and focuses on identifying and reducing risks before moving forward.

THADOMAL SHAHANI
TSEC
ENGINEERING COLLEGE

3] Incremental model:

The incremental model is a software process model where the system is developed and delivered in small, managable increments. Each increment adds new functionality to the previous release.

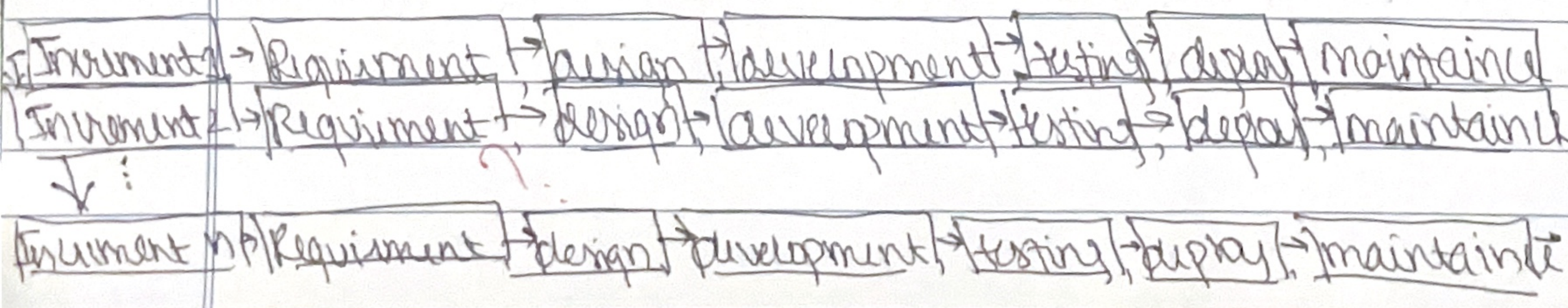
Phases in each increment:

1] Requirement Analysis
2] Design
3] Implementation
4] Testing
5] Deployment
6] Maintaince

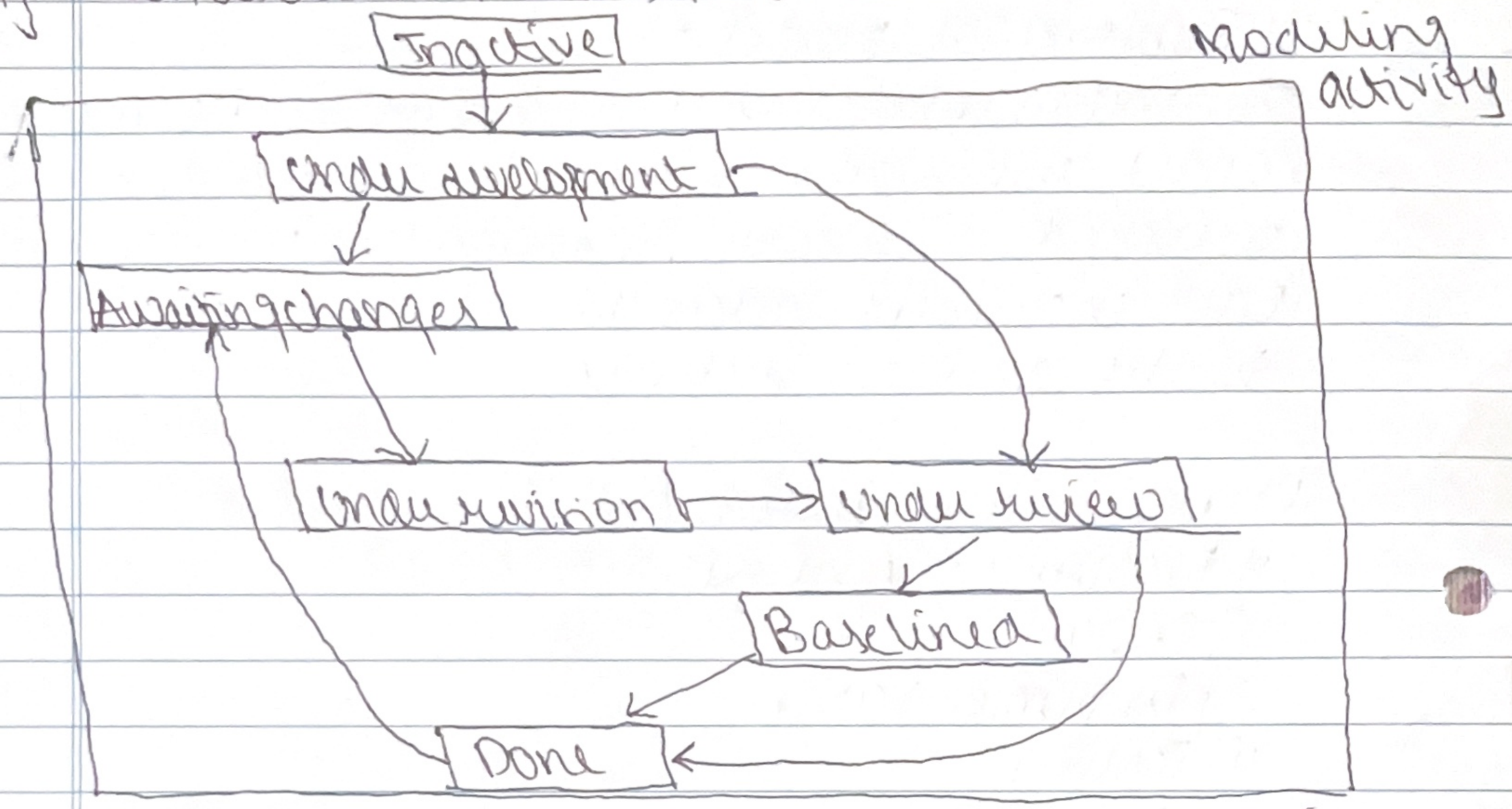
Increment 1 → Core Features
Increment 2 → Additional features
Increment 3 → Advanced features
⋮

Increment n → Final system.

In this:
Software is delivered in parts.
Each release is functional
Early ~~working~~ delivery of working software
User feedback is incorporated continously.



4] Concurrent Process Model:



This represents software development activities as states rather than rigid phases.
Multiple activities processing simultaneously.

- States:

1] Inactive → Activity has not started yet.
2] Under Development → Modeling work is currently being created.
3] Under review → Developed model is being reviewed.
4] Baselined → model is approved and placed under configuration control.
5] Done → Activity is completed successfully.
6] Awaiting changes → waiting for change requests or feedback.
7] Under Revision → Modifications are being made due to review comments or changes.